

Battery

Most smartphones these days come with Li-ion (lithium-ion) batteries. In the cell, Lithium ions move from the negative electrode to the positive electrode producing the current. These are re-chargeable but have a fixed number of life cycles. The capacity is measured in milliampere-hours (mAh). Phones with 1500 mAh battery rating will usually last long enough.



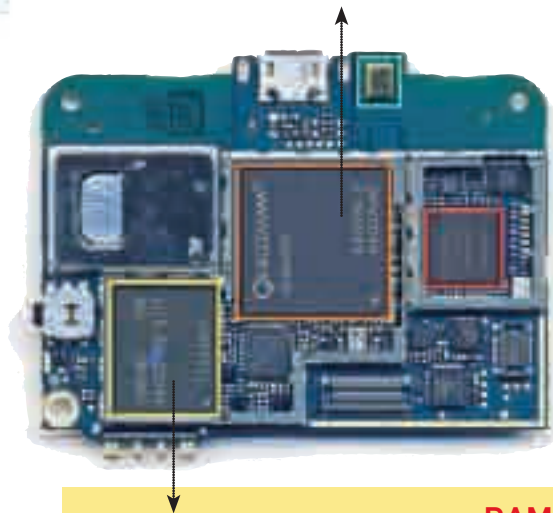
G-sensor

Smartphones sense their relative position or orientation by way of either an Accelerometer or Gyroscope. Most phones have Accelerometers to help keep track of orientation i.e. if you turn the phone it will ensure change from landscape to portrait and vice versa. This is done by analysing the vector acceleration which is caused as a result of weight (essentially g-force) experienced by a test mass inside the accelerometer device. Some high end smartphones such as iPhone 4G come with a Gyroscope which is capable of registering movement in a 3D space. So for instance it can detect the motion when you move the phone towards or away from your body. It only has applications in gaming as of now.

Processor

The processor is the heart or rather the brain of your smartphone; essentially what makes a smartphone smart. It does all the processing involved, be it when you run an application or even calculations to translate the position of your finger on the touch screen. Current high end smartphone processors are

clocked at 1GHz. Samsung, Qualcomm and Texas Instruments make these processors and are battling it out. Much like the desktop space, which is populated by names like Phenom, Core i and Athlon processors in the mobile space too at times acquire brand names. Samsung has Hummingbird, and Apple A4 while Qualcomm calls them Snapdragon, Scorpion etc; and Texas Instruments, OMAP. These chips are all based on processor cores from ARM, a company doesn't manufacture chips but rather only designs the cores for such application processors. Some of ARM's designs include older generation cores such as the ARM9, ARM11 and the relatively recent Cortex-A8 and Cortex-A9. Manufacturers make their own chipsets (System on Chip design) often combining a GPU, cellular modem, and GPS on board. The QSD8250 chipset featured here for instance, used the Adreno 200 graphics solution and a Cortex-A8 core, together called Snapdragon by Qualcomm.



Removable Storage

Removable storage for smartphones comes in the form of SD (secure digital) cards. Depending on size there are variants such as MicroSD (most common) and MiniSD. Phones can only support upto a certain amount of memory, with smartphones it usually 32GB. There are speed classes in SD cards specified by the SD association. Class 2 for example supports H.264 video recording, MPEG-4, and MPEG-2 video recording.

RAM

Memory in cellphones functions pretty much the same way it does in a desktop PC. The more the RAM, the more applications you can keep open, the more you can multitask. Low to mid range phones have RAM in the vicinity of 256MB, while high end phones such as the Desire HD have 768MB.